Test Case #1 Results:

```
Enter Reference String (e.g. 1 2 3 1 2): 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2
Enter Frame Count: 3

-----------------------------------------------------
| FIFO PAGE REPLACEMENT                             |
-----------------------------------------------------
Page    Frames      Result
7   7 - -       FAULT
0   7 0 -       FAULT
1   7 0 1       FAULT
2   2 0 1       FAULT
0   2 0 1       HIT
3   2 3 1       FAULT
0   2 3 0       FAULT
4   4 3 0       FAULT
2   4 2 0       FAULT
3   4 2 3       FAULT
0   0 2 3       FAULT
3   0 2 3       HIT
2   0 2 3       HIT

Total Page Faults: 10

-----------------------------------------------------
| LRU PAGE REPLACEMENT                              |
-----------------------------------------------------
Page    Frames      Result
7   7 - -       FAULT
0   7 0 -       FAULT
1   7 0 1       FAULT
2   0 1 2       FAULT
0   1 2 0       HIT
3   2 0 3       FAULT
0   2 3 0       HIT
4   3 0 4       FAULT
2   0 4 2       FAULT
3   4 2 3       FAULT
0   2 3 0       FAULT
3   2 0 3       HIT
2   0 3 2       HIT

Total Page Faults: 9

-----------------------------------------------------
| OPTIMAL PAGE REPLACEMENT                          |
-----------------------------------------------------
Page    Frames      Result
7   7 - -       FAULT
0   7 0 -       FAULT
1   7 0 1       FAULT
2   2 0 1       FAULT
0   2 0 1       HIT
3   2 0 3       FAULT
0   2 0 3       HIT
4   2 4 3       FAULT
2   2 4 3       HIT
3   2 4 3       HIT
0   2 0 3       FAULT
3   2 0 3       HIT
2   2 0 3       HIT

Total Page Faults: 7
```

Test Case #2 Results:

```
Enter Reference String (e.g. 1 2 3 1 2): 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5
Enter Frame Count: 3

-----------------------------------------------------
| FIFO PAGE REPLACEMENT                             |
-----------------------------------------------------
Page    Frames      Result
1   1 - -       FAULT
2   1 2 -       FAULT
3   1 2 3       FAULT
4   4 2 3       FAULT
1   4 1 3       FAULT
2   4 1 2       FAULT
5   5 1 2       FAULT
1   5 1 2       HIT
2   5 1 2       HIT
3   5 3 2       FAULT
4   5 3 4       FAULT
5   5 3 4       HIT

Total Page Faults: 9

-----------------------------------------------------
| LRU PAGE REPLACEMENT                              |
-----------------------------------------------------
Page    Frames      Result
1   1 - -       FAULT
2   1 2 -       FAULT
3   1 2 3       FAULT
4   2 3 4       FAULT
1   3 4 1       FAULT
2   4 1 2       FAULT
5   1 2 5       FAULT
1   2 5 1       HIT
2   5 1 2       HIT
3   1 2 3       FAULT
4   2 3 4       FAULT
5   3 4 5       FAULT

Total Page Faults: 10

-----------------------------------------------------
| OPTIMAL PAGE REPLACEMENT                          |
-----------------------------------------------------
Page    Frames      Result
1   1 - -       FAULT
2   1 2 -       FAULT
3   1 2 3       FAULT
4   1 2 4       FAULT
1   1 2 4       HIT
2   1 2 4       HIT
5   1 2 5       FAULT
1   1 2 5       HIT
2   1 2 5       HIT
3   3 2 5       FAULT
4   4 2 5       FAULT
5   4 2 5       HIT

Total Page Faults: 7
```

The Optimal Page Replacement in both test case 1 and 2 are indeed the same. As for the FIFO and LRU Page Replacements (in that order), the results for test case 1 are 10 and 9, and the results for test case 2 are 9 and 10.

Each algorithm, FIFO, LRU, and Optimal, has a different fault count primarily because they each count differently. FIFO means **F**irst **I**n **F**irst **O**ut, so pages that were entered into the reference earlier are also the first ones to be taken out. LRU removes pages that don't have much of a frequent occurrence. Then finally, Optimal removes any pages that the algorithm deems won't be necessary for the near future.